ELEC60023
ELEC70058

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2022

OPTOELECTRONICS

Thursday, 15 December 10:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible	First Marker(s) :	R.R.A. Syms
	Second Marker(s) :	O. Sydoruk

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Fundamental constants

$e = 1.6 \times 10^{-19}$ C

$m_e = 9.1 \times 10^{-31}$ kg

$\varepsilon_0 = 8.85 \times 10^{-12}$ F/m

$\mu_0 = 4\pi \times 10^{-7}$ m kg/C^2

$c = 3 \times 10^{8}$ m/s

$h = 6.62 \times 10^{-34}$ Js

Maxwell's equations – integral form, all fields time varying

$\iint_A \mathbf{D} \cdot d\underline{a} = \iiint_V \rho \, dv$

$\iint_A \mathbf{B} \cdot d\underline{a} = 0$

$\int_L \mathbf{E} \cdot d\underline{L} = - \iint_A \partial\mathbf{B}/\partial t \cdot d\underline{a}$

$\int_L \mathbf{H} \cdot d\underline{L} = \iint_A [\mathbf{J} + \partial\mathbf{D}/\partial t] \cdot d\underline{a}$

Maxwell's equations – differential form, all fields time varying

$\mathrm{div}(\mathbf{D}) = \rho$

$\mathrm{div}(\mathbf{B}) = 0$

$\mathrm{curl}(\mathbf{E}) = -\partial\mathbf{B}/\partial t$

$\mathrm{curl}(\mathbf{H}) = \mathbf{J} + \partial\mathbf{D}/\partial t$

Material or constitutive equations

$\mathbf{J} = \sigma \mathbf{E}$

$\mathbf{D} = \varepsilon \mathbf{E}$

$\mathbf{B} = \mu \mathbf{H}$

Vector calculus (Cartesian co-ordinates)

$\mathrm{grad}(\phi) = \partial\phi/\partial x \, \underline{i} + \partial\phi/\partial y \, \underline{j} + \partial\phi/\partial z \, \underline{k}$

$\mathrm{div}(\mathbf{F}) = \partial\mathbf{F}_x/\partial x + \partial\mathbf{F}_y/\partial y + \partial\mathbf{F}_z/\partial z$

$\mathrm{curl}(\mathbf{F}) = \underline{i} \{\partial\mathbf{F}_z/\partial y - \partial\mathbf{F}_y/\partial z\} + \underline{j} \{\partial\mathbf{F}_x/\partial z - \partial\mathbf{F}_z/\partial x\} + \underline{k} \{\partial\mathbf{F}_y/\partial x - \partial\mathbf{F}_x/\partial y\}$

$\mathrm{curl}\{\mathrm{curl}(\mathbf{F})\} = \mathrm{grad}\{\mathrm{div}(\mathbf{F})\} - \nabla^2\mathbf{F}$

$\iint_A \mathbf{F} \cdot d\underline{a} = \iiint_V \mathrm{div}(\mathbf{F}) \, dv$ (Gauss' Theorem)

$\int_L \mathbf{F} \cdot d\underline{L} = \iint_A \mathrm{curl}(\mathbf{F}) \cdot d\underline{a}$ (Stokes' Theorem)

1. a) Figure 1 shows two rays propagating in free space in the $x-z$ plane and representing plane waves of wavelength λ travelling at angles $\pm\theta$ to the z-axis. Assuming y-polarization, write down a time-independent expression for the electric field as a sum of two waves. Assuming the waves have equal amplitude, convert your solution into modal form.

[4]

b) Describe the modal solution, and identify the transverse field variation, the peak amplitude, the direction of travel and the effective propagation constant.

[4]

c) Identify the non-zero components of the magnetic field. Hence calculate the z-component of the irradiance and explain how an interference pattern will be formed on a screen perpendicular to the z-axis

[8]

d) Assuming the wavelength is $\lambda = 0.5\ \mu m$ and the angle is $\theta = 30^0$, find the period of the interference pattern.

[4]

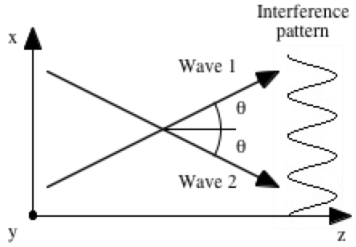


Figure 1.

2. a) A plane wave is incident on an interface between two media of refractive index n_1 and n_2. Sketch the outcome as a ray diagram, assuming the rays lie in the $x - z$ plane and the wave is incident from the high index side.

[4]

b) What relation must be satisfied between the angles of travel of the incident and transmitted wave? Explain the phenomenon of total internal reflection. What is the critical angle for an interface with $n_1 = 1.5$ and $n_2 = 1$?

[5]

c) Assuming all waves are y-polarized, write down an expression for the transmitted wave, i) before and ii) after the onset of total internal reflection. Sketch and explain the complete transverse field profile in case ii).

[6]

d) Figure 2 shows the spectral variation of loss for silica-based optical fibre. Identify the loss mechanisms in each of the spectral windows near A, B and C. What wavelength would be optimal for operation of the fibre in an optical communications system, and what would the loss be at this wavelength?

[5]

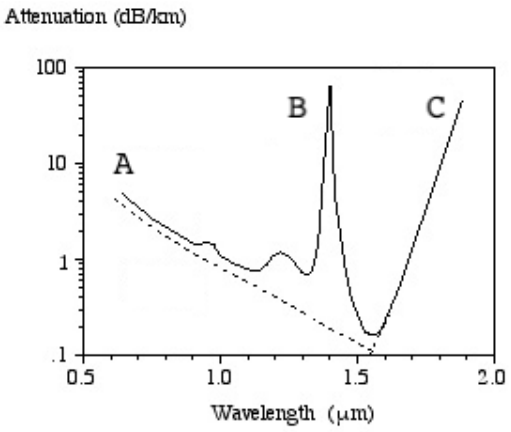


Figure 2.

3. a) Figure 3 shows an optical filter based on an unbalanced coupler-based Mach-Zehnder interferometer. Assuming the bar-state and cross-state outputs of a 3 dB coupler for unit input are $1/\sqrt{2}$ and $-j/\sqrt{2}$ and that the difference in length between the two arms is ΔL, derive expressions for the outputs as a function of wavelength λ.

[8]

b) Sketch the spectral responses predicted by your expression. Explain the disadvantages of MZI based filters in wavelength-division multiplexed optical communication systems.

[6]

c) Describe the advantageous properties of an arrayed waveguide grating multiplexer. Sketch the block diagram of an AWG MUX and explain how it may be combined with other components to form an ADD-DROP MUX.

[6]

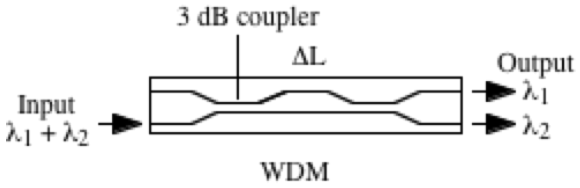


Figure 3.

4. The coupled mode equations for a synchronous directional coupler can be written as:

$$\frac{dA_1}{dx} + j\kappa A_2 = 0$$
$$\frac{dA_2}{dx} + j\kappa A_1 = 0$$

a) Explain the meaning of each term in the equations. What determines the value of κ, and how does κ vary with guide separation?

[4]

b) Solve the equations for unity input to Guide 1, and hence find the value of κL needed to form a 3 dB splitter in a device of length L.

[8]

c) If 3 dB splitters require a length $2\ mm$ for a guide separation of $3\ \mu m$ and a length $3\ mm$ for a separation of $4\ \mu m$, estimate the length required for a separation of $5\ \mu m$.

[8]

5. a) Figure 5 shows a photoconductive detector. Explain its operation.

[4]

b) Derive an expression for the spectral variation in responsivity and sketch the result.

[6]

c) Calculate the responsivity at a) $\lambda = 1\mu m$ and b) $\lambda = 2\mu m$, assuming the photoconductive material is silicon ($E_g = 1.14eV$) and the quantum efficiency is unity.

[4]

d) Briefly describe the drawbacks of photoconductive detectors and propose a solution in terms of photodiodes. On your previous diagram, superimpose the spectral variation of responsivity for a practical photodiode and explain any differences.

[6]

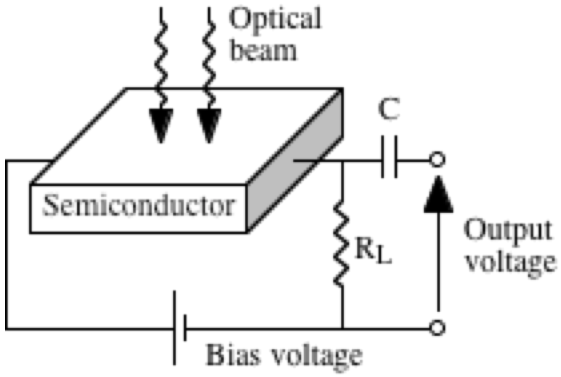


Figure 5.

6. The rate equations for a light-emitting diode can be written in the form:

$$\frac{dn}{dt} = \frac{I}{ev} - \frac{n}{\tau_e}$$
$$\frac{d\phi}{dt} = \frac{n}{\tau_{rr}} - \frac{\phi}{\tau_p}$$

a) Identify the physical processes described by the four blocks of terms on the right-hand sides of the equations. Why does the upper equation contain τ_e while the lower equation contains τ_{rr}? Why is the term ϕ/τ_p of interest, while ϕ itself is not?

[4]

b) Solve the equations in the steady state and hence find the internal efficiency η_i and the optical power output. Calculate η_i and the power output in mW/mA, for a GaAs LED emitting at $0.85\mu m$ wavelength, assuming that $\tau_{rr} = \tau_{nr} = 1ns$.

[6]

c) What factor limits the external efficiency η_e of an LED? Estimate the external efficiency of a GaAs LED, assuming the refractive index is 3.5.

[6]

d) Briefly describe the modifications needed to adapt LEDs that were originally developed for optical communications to lighting. What are the economic advantages and disadvantages of LED lighting?

[4]